



哈爾濱工業大學(深圳)

HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

Nonlinear and Adaptive Control

《非线性与自适应控制》

梅 杰

机电工程与自动化学院

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Introduction

Lecturer

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Introduction



群名称: 23秋非线性与自适应控制@HIT
群 号: 948663047



Course Objectives

1. Knowing the basic concepts and ideas of nonlinear and adaptive control, the applications in practice and research trends.
2. Familiar with the mainstream nonlinear control techniques and research directions, including **Lyapunov stability theory**, feedback linearization, backstepping, sliding mode control, model reference adaptive control, robust adaptive control, etc..
3. Using Matlab to design nonlinear and adaptive control algorithms and getting some simulation results.



Examination

Examination

Attendance: 10%

Homeworks: 20%

Final examination: 70%
closed-book, closed-note, closed-calculator exam

Final examination Time: TBD



- ◆ J-J E. Slotine and Weiping Li, *Applied Nonlinear Control*, English Edition, 机械工业出版社, 2004
Entry point for learning nonlinear control, minimization of math, readable and easy to understand
- ◆ Hassan K. Khalil, *Nonlinear Systems*, Third Edition, 电子工业出版社, 2007
Classic nonlinear control textbook, well-written and comprehensive, quite technical
- ◆ K.J. Åström and B. Wittenmark, *Adaptive Control*, Second Edition, Dover Publications, INC. Mineola, New York, 2008
- ◆ E. Lavertsky and K. Wise, *Robust and Adaptive Control with Aerospace Applications*, Springer-Verlag, 2013
- ◆ M. Krstic, I. Kanellakopoulos, and P. Kokotovic, *Nonlinear and Adaptive Control Design*, John Wiley and Sons, 1995.
- ◆ P. Ioannou and B. Fidan, *Adaptive Control Tutorial*, SIAM Press, Philadelphia, 2006
- ◆ Shankar Sastry, *Nonlinear Systems: Analysis, Stability, and Control*, 世界图书出版社, 1999
- ◆ K. Narendra and A. Annaswamy, *Stable Adaptive Systems*, Dover Publications, INC. Mineola, New York, 2005.



1. Introduction

what? why? how? and a brief history

2. Lyapunov Stability Theory: Autonomous Systems

Concepts of Stability, Linearization and Local Stability, the Invariance Principle..

3. Lyapunov Stability Theory: Non-autonomous Systems

Lyapunov-Like Analysis Using Barbalat's Lemma, LaSalle-Yoshizawa Theorem...

4. Advanced Stability Theory

BIBO Stability, Input-to-State Stability, ...

5. Adaptive Control

Model Reference Adaptive Control, Robust Adaptive Control

6. Nonlinear Control Design

Feedback Linearization, Backstepping, Sliding Mode Control,...

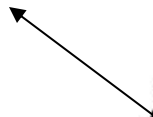
7. Control Applications

Control of Robotic Manipulators,...

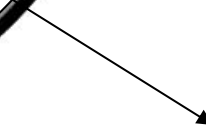


Linear Systems vs. Nonlinear Systems

Linear systems



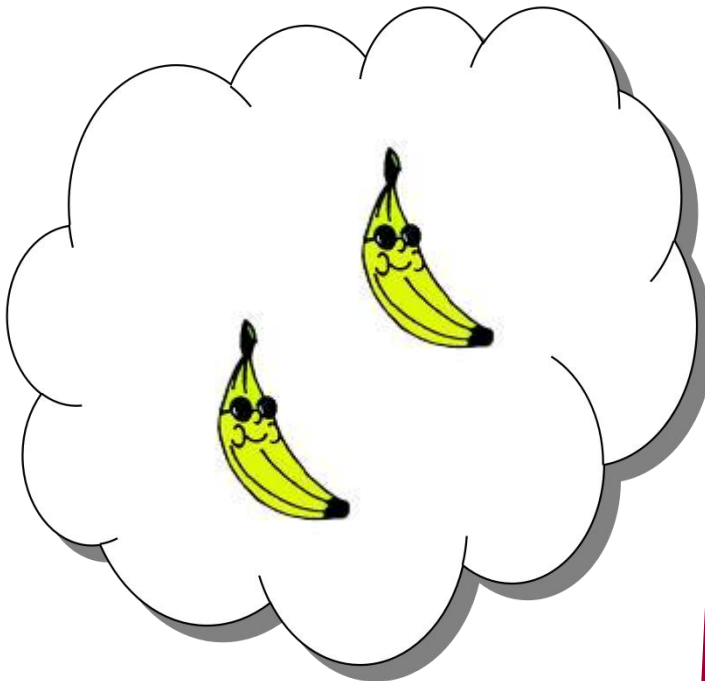
Nonlinear systems



Linear Systems vs. Nonlinear Systems

"Classifying systems as linear and nonlinear is like classifying the Universe as bananas and non-bananas" - unknown.

Linear systems



Nonlinear systems



Revisit of Linear Systems



System is linear if it satisfies

- ▶ Homogeneity. Given $y = \mathcal{G}(u)$, then

$$\alpha y = \mathcal{G}(\alpha u)$$

- ▶ Superposition. If

$$y_1 = \mathcal{G}(u_1) \quad y_2 = \mathcal{G}(u_2)$$

then

$$y_1 + y_2 = \mathcal{G}(u_1 + u_2)$$

System is nonlinear if it is not linear.



- LTI system expressible as ODE with constant coefficients

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y^{(1)} + a_n y = b_0 u^{(m)} + b_1 u^{(m-1)} + \dots + b_m u$$

- Transfer function representation

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^m + b_1 s^{(m-1)} + \dots + b_m}{s^n + a_1 s^{(n-1)} + \dots + a_{n-1} s + a_n}$$

- State Space Representation

$$\begin{aligned}\dot{\mathbf{x}} &= A\mathbf{x} + B\mathbf{u} & \mathbf{x}(0) &= \mathbf{x}_0 \\ \mathbf{y} &= C\mathbf{x} + D\mathbf{u}\end{aligned}$$

➤ Classic approach (Transfer function representation):

- Laplace transform
- Routh-Hurwitz criterion
- Root locus
- Bode diagram
- Nyquist criterion

➤ Modern approach (State space representation):

- Pole placement
- Observe design
- Internal model design
- Optimal control



Why Nonlinear Control?

- All systems are nonlinear -- nonlinear control extends range of possible operation and linear systems are not rich enough to describe many commonly observed phenomena
- Some hardware nonlinearities do not have linear approximations
- System uncertainty should be treated with nonlinear control (**adaptive control**)
- Some nonlinear control designs based on physical principles -- provide better intuition and understanding
- Better cost/performance than assembly of local linear controls
- Empowered by modern computational tools

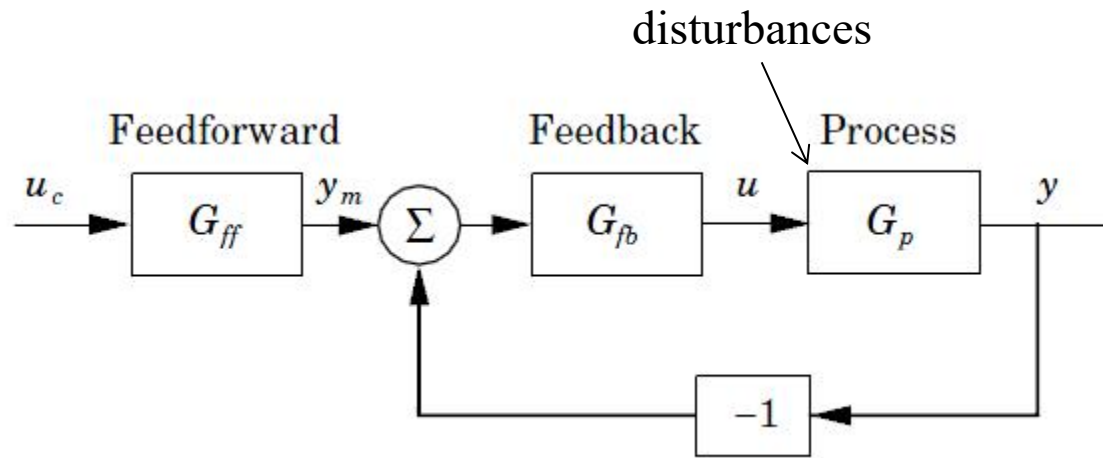


A Narrow Definition of Adaptive Control

- An adaptive controller is a **fixed-structure** controller with **adjustable parameters** and a mechanism for automatically adjusting those parameters
- In this sense, an adaptive controller is one way of dealing with **parametric** uncertainty
- Adaptive control theory essentially deals with finding parameter adjustment algorithms that guarantee stability and convergence

Robust High-Gain Control

Feedback by itself has the ability to cope with **parameter changes and uncertainties**.



Block diagram of a robust high-gain system

G_{fb} : **attenuate disturbances** and make the closed-loop system **insensitive to process variations**

G_{ff} : give the desired response to command signals

The transfer function from y_m to y is

$$T = \frac{G_p G_{fb}}{1 + G_p G_{fb}}$$

Taking derivatives with respect to G_p , we get

$$\frac{dT}{T} = \frac{1}{1 + G_p G_{fb}} \frac{dG_p}{G_p}$$

the loop transfer function $L = G_p G_{fb}$

A blue arrow points from the term $G_p G_{fb}$ in the denominator of the derivative equation above to the definition of the loop transfer function $L = G_p G_{fb}$.

To design a robust controller, it is thus attempted to find G_{fb} such that the loop transfer function is large for those frequencies at which there are large variations in the process transfer function.



Example 1.1 Different Open-Loop Responses

Consider systems with the open-loop transfer functions

$$G_0(s) = \frac{1}{(s+1)(s+a)}$$

where $a = -0.01, 0$, and 0.01 .

The system with $a = 0.01$ is stable; the others are unstable.



Example 1.1 Different Open-Loop Responses

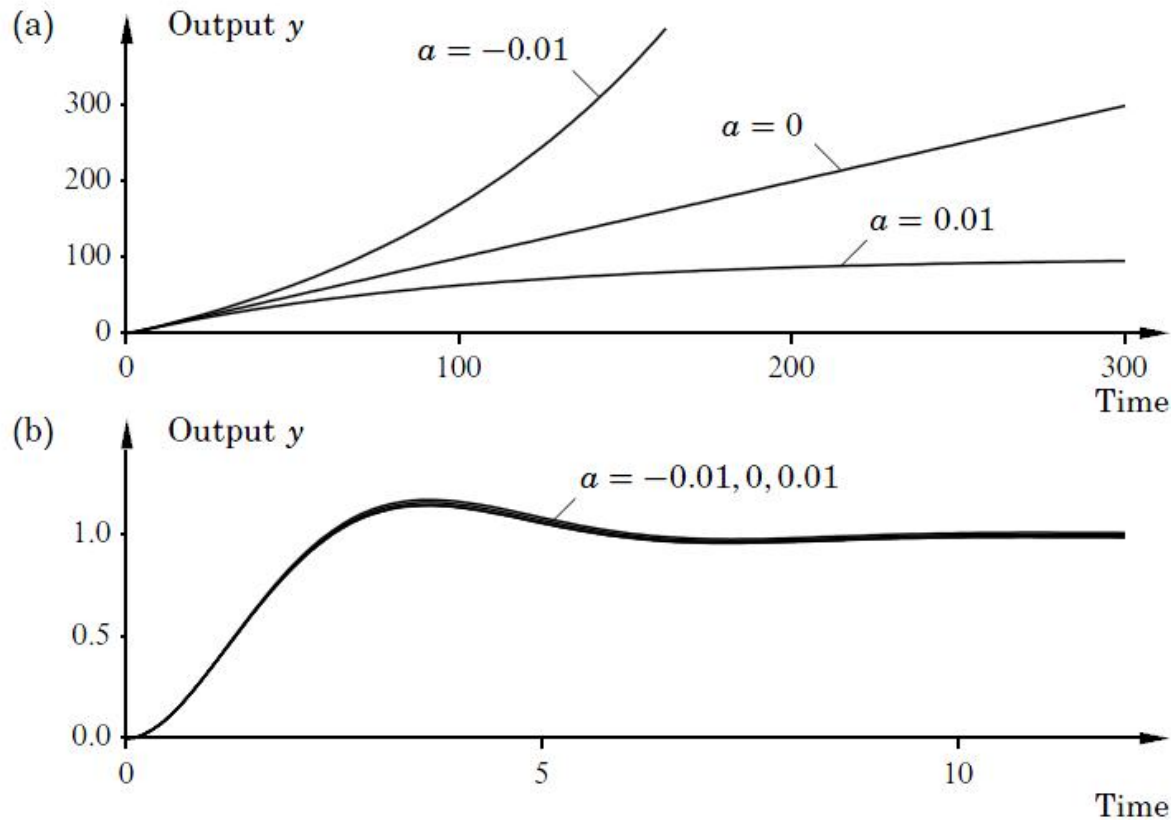


Figure 1.4 (a) Open-loop unit step responses for the process in Example 1.1 with $\alpha = -0.01, 0, 0.01$. (b) Closed-loop unit step responses for the same system with the same α values. The difference in time scales.

上图：开环响应(根本没有效果)
下图：闭环响应(效果明显更好)



Example 1.2 Similar Open-Loop Responses

Consider systems with the open-loop transfer functions

$$G_0(s) = \frac{400(1 - sT)}{(s + 1)(s + 20)(1 + Ts)}$$

with $T = 0, 0.015$, and 0.03 .

The transfer functions are similar for different T s.



Example 1.2 Similar Open-Loop Responses

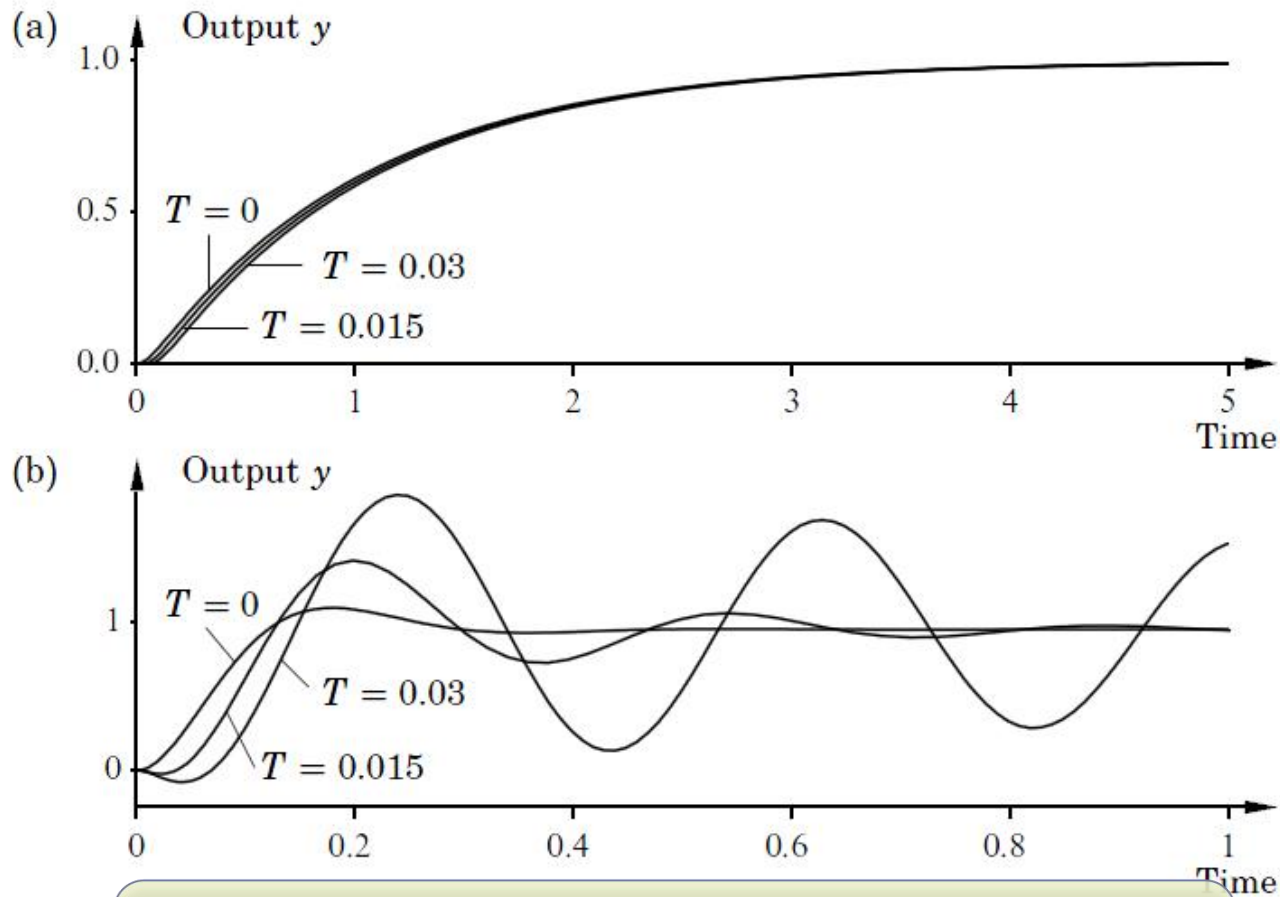


Figure 1.6

上图：开环响应(效果很好)

下图：闭环响应(效果反而比较差?)

with the feedback $u = u_c - y$. Notice the difference in time scales.



Example 1.3 Integrator with Unknown Gain

Consider a process whose dynamics is described by

$$G_0(s) = \frac{k_p}{s}$$

where the gain k_p can assume both positive and negative values.

This is a very severe variation because the phase of the system can change by 180°.

Let the controller transfer function be $S(s)/R(s)$, where $R(s)$ and $S(s)$ are polynomials.

The characteristic polynomial of the closed-loop system is then

$$P(s) = sR(s) + k_p S(s)$$

A necessary condition for $P(s)$ to have all roots in the left half-plane is that all coefficients are positive.



Examples 1.1-1.3 -Summary

- Parameter variations may result in significantly difference open-loop responses.
- Significant variations in open-loop responses may have litter effect on the closed-loop performances via feedback control; However, there are also situations that the closed-loop responses may be significantly different even if changes in the open-loop responses are small.
- Some process cannot be controlled by a linear feedback controller.



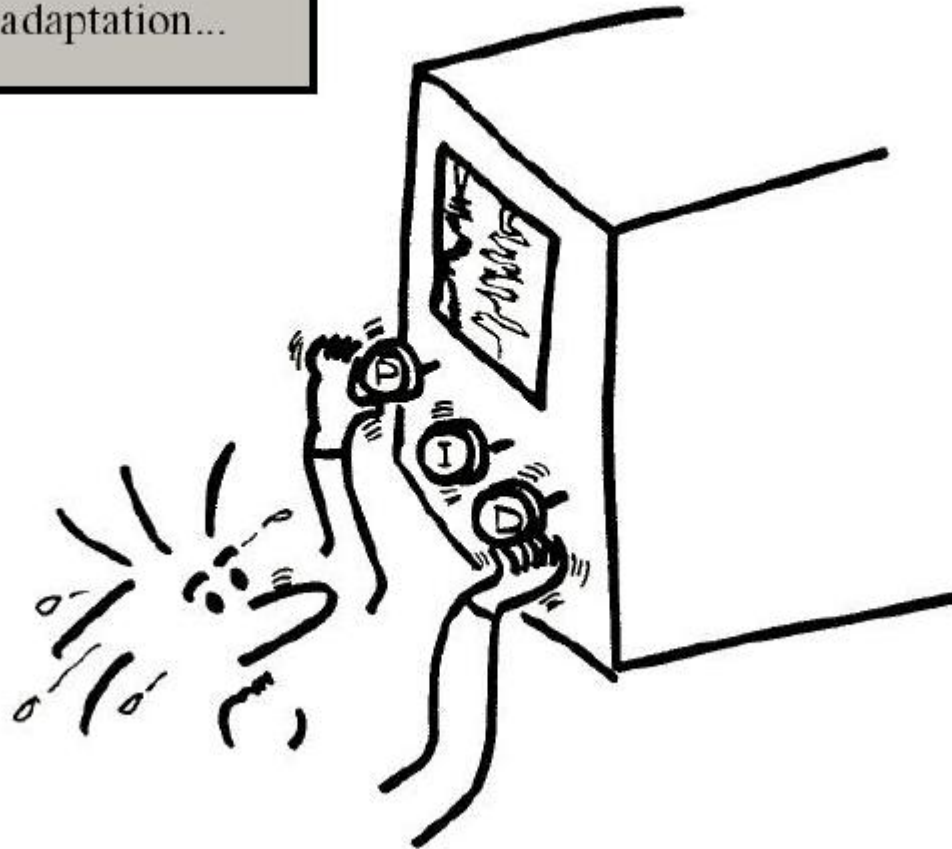
Why Adaptive Control?

- Systems to be controlled have **parameter uncertainty**
 - unknown mass, length of links, and the inertial parameters of the loads in robot manipulation;
 - slowly varying mass of a spacecraft.
- Systems experience **unpredictable parameter variations**
 - The dynamic behavior of an aircraft depends on its altitude, speed, and configuration;
- Unknown **disturbance characteristics**

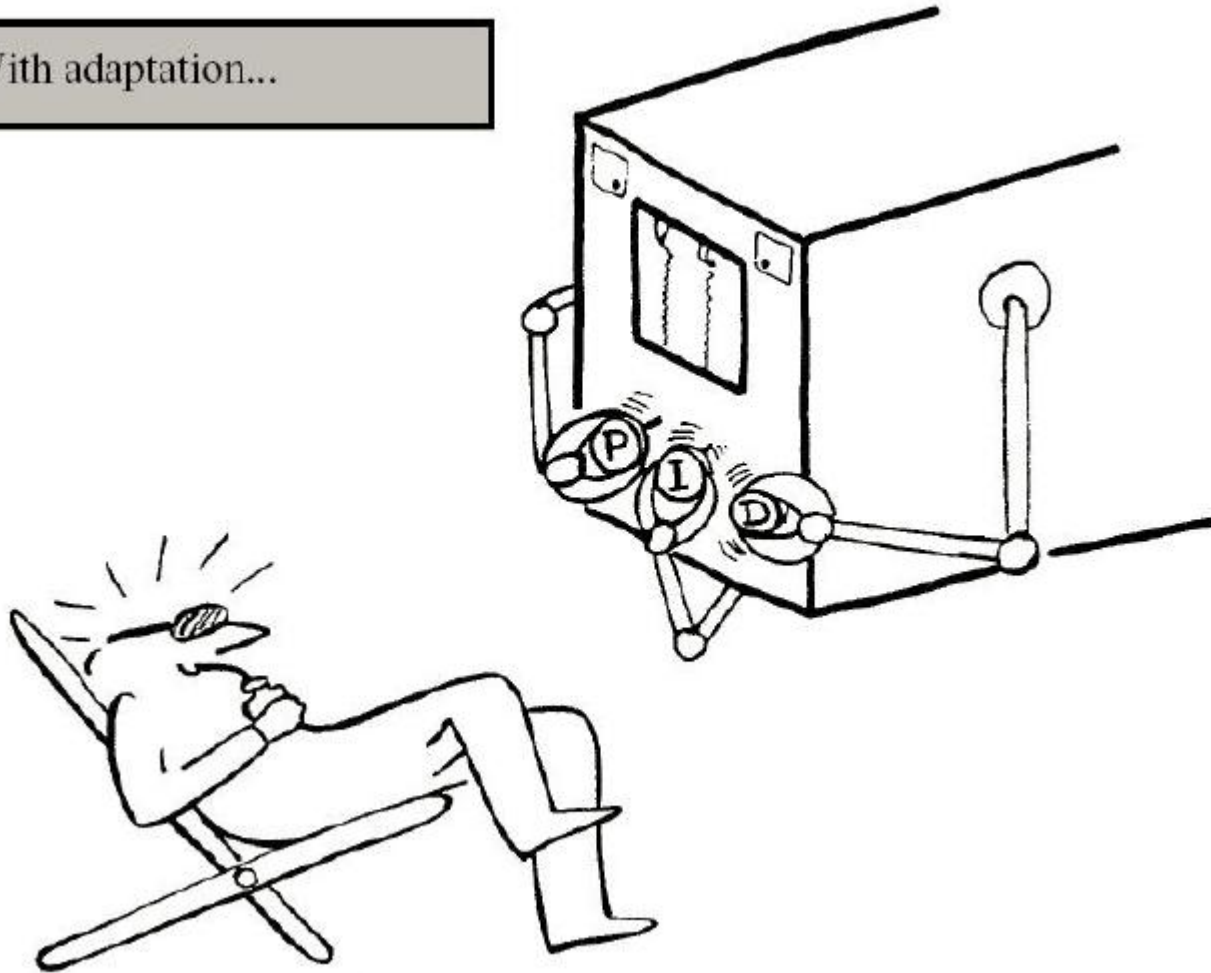
Basic Objective

Maintain consistent performance of a system in the presence of uncertainties and variations in plant parameters.

Without adaptation...



With adaptation...





航天控制未来发展的思考

基于能力自适应协调的质心/绕心一体化控制理论

近空间复杂空域长时间高速飞行，非定常复杂流场容易诱发面对称飞行器产生横侧向耦合、气动控制耦合等，需要解决飞行器控制能力的自适应协调、基于能力的控制策略选择问题。

关键技术：质心/绕心运动自适应协调策略、横侧向稳定性协调控制、非定常流场脉动压力耦合共振控制等。



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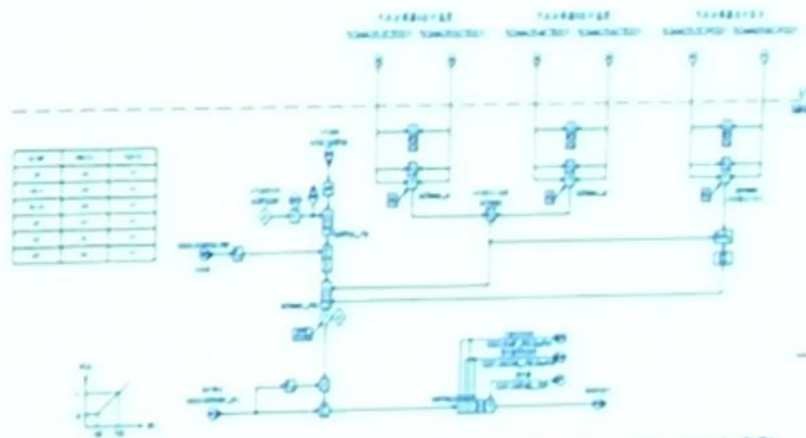




4.2 超/超超临界直流锅炉高安全CCS

——主蒸汽压力控制新方法

- 研制出专用液位/压力/温度变送器，提出根据实际压力与压力设定值偏差为前馈参数的主蒸汽压力控制新方法DEB，构建了一整套前馈控制参数自适应更新规则，有效地减小了负荷大范围变动下的控制品质



超临界600MW机组锅炉燃烧率需求计算

- ✓突破传统前馈方法参数固定且未考虑变负荷过程中实际压力与设定值偏差的不足
- ✓高负荷段相对加强燃料与给水的前馈超前指令强度

有效解决负荷动态变化条件下主蒸汽压力控制的技术难题



Applications

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08:30-09:30 Tulum Ballroom A-H Plenary Session TuPL Contraction Theory in Systems and Control										
09:30-10:00 Room T19 TuAM_BR										
10:00-12:00 Tulum Ballroom B Regular Session TuAT02 Adaptive Control I	10:00-12:00 Tulum Ballroom C Regular Session TuAT03 Robotics I	10:00-12:00 Tulum Ballroom D Regular Session TuAT04 Machine Learning I	10:00-12:00 Tulum Ballroom E Invited Session TuAT05 Risk-Aware Learning, Verification, and Control	10:00-12:00 Tulum Ballroom F Regular Session TuAT06 Kernel-Based Identification	10:00-12:00 Tulum Ballroom G Regular Session TuAT07 Estimation	10:00-12:00 Tulum Ballroom H Regular Session TuAT08 Alternating Direction Method of Multipliers	10:00-12:00 Maya Ballroom I Regular Session TuAT09 Consensus of Multi-Agent Systems	10:00-12:00 Maya Ballroom II Regular Session TuAT10 Stochastic Systems I	10:00-12:00 Maya Ballroom III Regular Session TuAT11 Distributed Parameter Systems I	10:00-12:00 Maya Ballroom IV Invited Session TuAT12 Event-Triggered Control
12:00-13:30 Room T19 TuLU_BR										
13:30-15:30 Tulum Ballroom B Regular Session TuBT02 Adaptive Control II	13:30-15:30 Tulum Ballroom C Regular Session TuBT03 Robotics II	13:30-15:30 Tulum Ballroom D Regular Session TuBT04 Machine Learning II	13:30-15:30 Tulum Ballroom E Invited Session TuBT05 Control and Estimation of Traffic Flow Systems	13:30-15:30 Tulum Ballroom F Regular Session TuBT06 Identification of Linear Systems	13:30-15:30 Tulum Ballroom G Regular Session TuBT07 Fault Tolerant Systems I	13:30-15:30 Tulum Ballroom H Regular Session TuBT08 Barrier Functions in Constrained Control	13:30-15:30 Maya Ballroom I Regular Session TuBT09 Network Analysis and Control I	13:30-15:30 Maya Ballroom II Regular Session TuBT10 Stochastic Systems II	13:30-15:30 Maya Ballroom III Regular Session TuBT11 Distributed Parameter Systems II	13:30-15:30 Maya Ballroom IV Tutorial Session TuBT12 Recent Advances in Data-Driven Control: Concepts, Theory, and Applications
15:30-16:00 Room T19 TuPM_BR										
16:00-18:00 Tulum Ballroom B Regular Session TuCT02 Adaptive Control III	16:00-18:00 Tulum Ballroom C Regular Session TuCT03 Robotics III	16:00-18:00 Tulum Ballroom D Regular Session TuCT04 Learning	16:00-18:00 Tulum Ballroom E Regular Session TuCT05 Iterative Learning Control	16:00-18:00 Tulum Ballroom F Regular Session TuCT06 Neural Networks for Identification	16:00-18:00 Tulum Ballroom G Regular Session TuCT07 Fault Tolerant Systems II	16:00-18:00 Tulum Ballroom H Regular Session TuCT08 Learning and Optimization	16:00-18:00 Maya Ballroom I Regular Session TuCT09 Network Analysis and Control II	16:00-18:00 Maya Ballroom II Regular Session TuCT10 Stochastic Systems III	16:00-18:00 Maya Ballroom III Regular Session TuCT11 Robust Control I	16:00-18:00 Maya Ballroom IV Invited Session TuCT12 Event-Triggered Control for Multi-Vehicle, Multi-Robot, and Multi-Agent Systems





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08:30-09:30 Tulum Ballroom A-H Plenary Session WePL											
How Do We Learn to Use Learning in Manufacturing Systems											
09:30-10:00 Room T19 WeAM_BR											
10:00-12:00 Tulum Ballroom A Regular Session WeAT01 Hybrid Systems I	10:00-12:00 Tulum Ballroom B Regular Session WeAT02 Adaptive Control IV	10:00-12:00 Tulum Ballroom C Regular Session WeAT03 Robotics IV	10:00-12:00 Tulum Ballroom D Regular Session WeAT04 Neural Networks I	10:00-12:00 Tulum Ballroom E Invited Session WeAT05 Learning-Based Control I: Data-Driven Predictive Control	10:00-12:00 Tulum Ballroom F Regular Session WeAT06 Parameter Identification	10:00-12:00 Tulum Ballroom G Invited Session WeAT07 Modular Design and Verification of Control Systems	10:00-12:00 Tulum Ballroom H Regular Session WeAT08 Resilient Control Systems	10:00-12:00 Maya Ballroom I Regular Session WeAT09 Cooperative Control	10:00-12:00 Maya Ballroom II Regular Session WeAT10 Stochastic Systems IV	10:00-12:00 Maya Ballroom III Regular Session WeAT11 Robust Control II	10:00-12:00 Maya Ballroom IV Invited Session WeAT12 Learning and Resilience in Event-Triggered Control
12:00-13:30 Room T19 WeLU_BR											
13:30-15:30 Tulum Ballroom A Regular Session WeBT01 Hybrid Systems I	13:30-15:30 Tulum Ballroom B Regular Session WeBT02 Adaptive Systems	13:30-15:30 Tulum Ballroom C Regular Session WeBT03 Robotics V	13:30-15:30 Tulum Ballroom D Regular Session WeBT04 Neural Networks II	13:30-15:30 Tulum Ballroom E Invited Session WeBT05 Learning-Based Control II: Analysis, Estimation and Control	13:30-15:30 Tulum Ballroom F Regular Session WeBT06 Identification for Control	13:30-15:30 Tulum Ballroom G Invited Session WeBT07 Online Learning, Optimization, and Game Theory I	13:30-15:30 Tulum Ballroom H Regular Session WeBT08 Formal Verification and Synthesis	13:30-15:30 Maya Ballroom I Regular Session WeBT09 Networked Control Systems I	13:30-15:30 Maya Ballroom II Regular Session WeBT10 Stochastic Optimal Control I	13:30-15:30 Maya Ballroom III Regular Session WeBT11 Robust Control III	13:30-15:30 Maya Ballroom IV Tutorial Session WeBT12 Cybersecurity and Supervisory Control: Synthesis of Attacks and of Defense Strategies Using Discrete Event Systems Techniques
15:30-16:00 Room T19 WePM_BR											
16:00-18:00 Tulum Ballroom A Regular Session WeCT01 Hybrid Systems III	16:00-18:00 Tulum Ballroom B Regular Session WeCT02 Linear Parameter-Varying Systems	16:00-18:00 Tulum Ballroom C Regular Session WeCT03 Autonomous Robots	16:00-18:00 Tulum Ballroom D Regular Session WeCT04 Neural Networks III	16:00-18:00 Tulum Ballroom E Invited Session WeCT05 Learning-Based Control III: Robustness and Safety	16:00-18:00 Tulum Ballroom F Regular Session WeCT06 Estimation and Input Design	16:00-18:00 Tulum Ballroom G Invited Session WeCT07 Online Learning, Optimization, and Game Theory II	16:00-18:00 Tulum Ballroom H Regular Session WeCT08 Social and Financial Networks	16:00-18:00 Maya Ballroom I Regular Session WeCT09 Networked Control Systems II	16:00-18:00 Maya Ballroom II Regular Session WeCT10 Stochastic Optimal Control II	16:00-18:00 Maya Ballroom III Regular Session WeCT11 Robust Control IV	16:00-18:00 Maya Ballroom IV Regular Session WeCT12 Sampled-Data Control and Estimation





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08:30-09:30 Tulum Ballroom A-H Plenary Session ThPL											
Nonlinear Systems with Limited Data: Estimation, Control and Synchronization											
09:30-10:00 Room T19 ThAM_BR											
10:00-12:00 Tulum Ballroom G Invited Session ThAT07	10:00-12:00 Tulum Ballroom H Invited Session ThAT08	10:00-12:00 Maya Ballroom I Regular Session ThAT09	10:00-12:00 Maya Ballroom II Regular Session ThAT10	10:00-12:00 Maya Ballroom III Regular Session ThAT11	10:00-12:00 Maya Ballroom IV Invited Session ThAT12	10:00-12:00 Maya Ballroom V Regular Session ThAT13	10:00-12:00 Maya Ballroom VI Regular Session ThAT14	10:00-12:00 Maya Ballroom VII Regular Session ThAT15	10:00-12:00 Maya Ballroom VIII Regular Session ThAT16	10:00-12:00 Acapulco Regular Session ThAT17	
Multi-Agent Optimization and Games	Robust Distributed Optimization, Estimation, and Coordination in Multi-Agent Systems	Linear Systems I	Discrete-Event Systems	Sliding Mode Control	Modeling, Estimation, and Control of Epidemic Processes	Constrained Control and Optimization I	Energy Systems	Game Theory IV	Nonlinear Output Feedback	Optimal Control and Applications	
12:00-13:30 Room T19 ThLU_BR											
13:30-15:30 Tulum Ballroom G Invited Session ThBT07	13:30-15:30 Tulum Ballroom H Regular Session ThBT08	13:30-15:30 Maya Ballroom I Regular Session ThBT09	13:30-15:30 Maya Ballroom II Regular Session ThBT10	13:30-15:30 Maya Ballroom III Regular Session ThBT11	13:30-15:30 Maya Ballroom IV Tutorial Session ThBT12	13:30-15:30 Maya Ballroom V Regular Session ThBT13	13:30-15:30 Maya Ballroom VI Regular Session ThBT14	13:30-15:30 Maya Ballroom VII Regular Session ThBT15	13:30-15:30 Maya Ballroom VIII Regular Session ThBT16	13:30-15:30 Acapulco Regular Session ThBT17	
Distributionally Robust Optimization and Control	Secure Control Systems	Linear Systems II	Markov Processes	Uncertain Systems I	Modelling and Control of Epidemics across Scales	Constrained Control and Optimization II	Dimensioning and Operation of Energy Systems	Game Theory V	Lyapunov Methods I	Quantum Information and Control I	
15:30-16:00 Room T19 ThPM_BR											
16:00-18:00 Tulum Ballroom G Invited Session ThCT07	16:00-18:00 Tulum Ballroom H Regular Session ThCT08	16:00-18:00 Maya Ballroom I Regular Session ThCT09	16:00-18:00 Maya Ballroom II Regular Session ThCT10	16:00-18:00 Maya Ballroom III Regular Session ThCT11	16:00-18:00 Maya Ballroom IV Regular Session ThCT12	16:00-18:00 Maya Ballroom V Regular Session ThCT13	16:00-18:00 Maya Ballroom VI Invited Session ThCT14	16:00-18:00 Maya Ballroom VII Regular Session ThCT15	16:00-18:00 Maya Ballroom VIII Regular Session ThCT16	16:00-18:00 Acapulco Regular Session ThCT17	
Estimation and Control of Infinite-Dimensional Systems I	Privacy and Security	Stability of Linear Systems	Mean Field Games	Uncertain Systems II	Modeling and Control of Epidemics	Applications of Optimization	Trends in Optimization for Power Systems	Game-Theoretic Methods in Agent-Based Systems	Lyapunov Methods II	Quantum Information and Control II	





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09:30-10:00 Room T19 FrAM_BR										
10:00-12:00 Tulum Ballroom G Invited Session FrAT07 Estimation and Control of Infinite-Dimensional Systems II	10:00-12:00 Tulum Ballroom H Invited Session FrAT08 Resilience, Security, and Privacy for Intelligent and Interconnected Systems I	10:00-12:00 Maya Ballroom I Regular Session FrAT09 Linear Optimal Control	10:00-12:00 Maya Ballroom II Invited Session FrAT10 Control-Theoretic Perspectives on Optimization Algorithms	10:00-12:00 Maya Ballroom III Regular Session FrAT11 Delay Systems	10:00-12:00 Maya Ballroom IV Invited Session FrAT12 Management and Control Strategies for Emerging Mobility Systems in Smart Cities	10:00-12:00 Maya Ballroom V Regular Session FrAT13 Computational Methods and Optimization	10:00-12:00 Maya Ballroom VI Invited Session FrAT14 Control for Energy and Climate Security	10:00-12:00 Maya Ballroom VII Invited Session FrAT15 Distributed Game Equilibrium Seeking Methods	10:00-12:00 Maya Ballroom VIII Regular Session FrAT16 Stability of Nonlinear Systems I	10:00-12:00 Acapulco Invited Session FrAT17 Resilience and Robustness in Large-Scale Networked Dynamical Systems
12:00-13:30 Room T19 FrLU_BR										
13:30-15:30 Tulum Ballroom G Invited Session FrBT07 Estimation and Control of Infinite-Dimensional Systems III	13:30-15:30 Tulum Ballroom H Invited Session FrBT08 Resilience, Security, and Privacy for Intelligent and Interconnected Systems II	13:30-15:30 Maya Ballroom I Regular Session FrBT09 Optimal Control Algorithms	13:30-15:30 Maya Ballroom II Regular Session FrBT10 Optimization I	13:30-15:30 Maya Ballroom III Invited Session FrBT11 Stability and Stabilization of Time-Delay Systems: Recent Trends and Applications	13:30-15:30 Maya Ballroom IV Tutorial Session FrBT12 New Frontiers of Traffic Control and Estimation	13:30-15:30 Maya Ballroom V Regular Session FrBT13 Optimization Algorithms I	13:30-15:30 Maya Ballroom VI Regular Session FrBT14 Vehicular Technology	13:30-15:30 Maya Ballroom VII Invited Session FrBT15 Dynamics on Complex Networks: Estimation, Analysis and Control	13:30-15:30 Maya Ballroom VIII Regular Session FrBT16 Stability of Nonlinear Systems II	13:30-15:30 Acapulco Invited Session FrBT17 Learning and AI Methods for CPS Security
15:30-16:00 Room T19 FrPM_BR										
16:00-18:00 Tulum Ballroom G Invited Session FrCT07 Estimation and Control of Infinite-Dimensional Systems IV	16:00-18:00 Tulum Ballroom H Invited Session FrCT08 Networked CPS Resilience	16:00-18:00 Maya Ballroom I Regular Session FrCT09 Nonlinear Optimal Control	16:00-18:00 Maya Ballroom II Regular Session FrCT10 Optimization II	16:00-18:00 Maya Ballroom III Regular Session FrCT11 Large-Scale Systems	16:00-18:00 Maya Ballroom IV Invited Session FrCT12 Dynamic Games and Networked Systems	16:00-18:00 Maya Ballroom V Regular Session FrCT13 Optimization Algorithms II	16:00-18:00 Maya Ballroom VI Regular Session FrCT14 Mechatronics	16:00-18:00 Maya Ballroom VII Invited Session FrCT15 Dealing with Constraints in Networked Control Systems	16:00-18:00 Maya Ballroom VIII Regular Session FrCT16 Stability of Nonlinear Systems III	16:00-18:00 Acapulco Invited Session FrCT17 Encrypted Control and Optimization





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	Track 13	Track 14	Track 15	Track 16	Track 17	Track 18	Track 19	Track 20	Track 21
	08:30-09:30 Orchid Main 4202-4306 Semipenary Session WeSP2 <i>Fast and Flexible Multi-Agent Decision-Making</i>								
10:00-12:00	Roselle Junior 4613 Invited Session WeA13 <i>Identification in Networked Systems</i>	Roselle Junior 4612 Invited Session WeA14 <i>Modeling, Analysis, and Control of Complex Systems</i>	Roselle Junior 4611 Regular Session WeA15 <i>Adaptive Control I</i>	Peony Junior 4512 Invited Session WeA16 <i>Computational Techniques for Automation in Energy Systems</i>	Peony Junior 4511 Regular Session WeA17 <i>Data-Driven Control I</i>	Peony Junior 4412 Regular Session WeA18 <i>Nonlinear Systems I</i>	Peony Junior 4411 Regular Session WeA19 <i>Linear Systems I</i>	Orchid Junior 4312 Regular Session WeA20 <i>Biological Systems I</i>	Orchid Junior 4311 Regular Session WeA21 <i>Constrained Control I</i>
13:30-15:30	Roselle Junior 4613 Regular Session WeB13 <i>Control of Networks</i>	Roselle Junior 4612 Regular Session WeB14 <i>Identification II</i>	Roselle Junior 4611 Regular Session WeB15 <i>Adaptive Control II</i>	Peony Junior 4512 Regular Session WeB16 <i>Energy Systems I</i>	Peony Junior 4511 Regular Session WeB17 <i>Data-Driven Control II</i>	Peony Junior 4412 Regular Session WeB18 <i>Nonlinear Systems II</i>	Peony Junior 4411 Regular Session WeB19 <i>Linear Systems II</i>	Orchid Junior 4312 Regular Session WeB20 <i>Biological Systems II</i>	Orchid Junior 4311 Regular Session WeB21 <i>Constrained Control II</i>
16:00-18:00	Roselle Junior 4613 Regular Session WeC13 <i>Network Analysis</i>	Roselle Junior 4612 Regular Session WeC14 <i>Identification III</i>	Roselle Junior 4611 Regular Session WeC15 <i>Adaptive Control III</i>	Peony Junior 4512 Regular Session WeC16 <i>Energy Systems II</i>	Peony Junior 4511 Regular Session WeC17 <i>Data-Driven Control III</i>	Peony Junior 4412 Regular Session WeC18 <i>Nonlinear Systems III</i>	Peony Junior 4411 Regular Session WeC19 <i>Linear Systems III</i>	Orchid Junior 4312 Regular Session WeC20 <i>Biomolecular Systems</i>	Orchid Junior 4311 Regular Session WeC21 <i>Constrained Control III</i>





Applications

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Click on the day's program and use the arrow keys for easy horizontal and vertical scrolling

CDC 2023 Technical Program Thursday December 14, 2023									
Track 12	Track 13	Track 14	Track 15	Track 16	Track 17	Track 18	Track 19	Track 20	Track 21
08:30-09:30 Orchid Main 4202-4306 Semiplenary Session ThSP2 Dual Control Revisited									
10:00-12:00 Roselle Junior 4711 Regular Session ThA12 Operative Control IV	10:00-12:00 Roselle Junior 4613 Regular Session ThA13 Networked Control Systems I	10:00-12:00 Roselle Junior 4612 Regular Session ThA14 Identification II	10:00-12:00 Roselle Junior 4611 Regular Session ThA15 Robust Adaptive Control	10:00-12:00 Peony Junior 4512 Regular Session ThA16 Power Systems I	10:00-12:00 Peony Junior 4511 Invited Session ThA17 Inverse Problems in Control, Estimation and Reinforcement Learning	10:00-12:00 Peony Junior 4412 Regular Session ThA18 Nonlinear Systems IV	10:00-12:00 Peony Junior 4411 Regular Session ThA19 Linear Parameter-Varying Systems	10:00-12:00 Orchid Junior 4312 Invited Session ThA20 Stochastic and Distributed Models in Systems and Synthetic Biology	10:00-12:00 Orchid Junior 4311 Regular Session ThA21 Predictive Control for Nonlinear Systems I
13:30-15:30 Roselle Junior 4711 Regular Session ThB12 Operative Control V	13:30-15:30 Roselle Junior 4613 Regular Session ThB13 Networked Control Systems II	13:30-15:30 Roselle Junior 4612 Regular Session ThB14 Identification IV	13:30-15:30 Roselle Junior 4611 Regular Session ThB15 Robust Control I	13:30-15:30 Peony Junior 4512 Regular Session ThB16 Power Systems II	13:30-15:30 Peony Junior 4511 Regular Session ThB17 Iterative Learning Control I	13:30-15:30 Peony Junior 4412 Regular Session ThB18 Stability of Nonlinear Systems I	13:30-15:30 Peony Junior 4411 Regular Session ThB19 Stability of Linear Systems	13:30-15:30 Orchid Junior 4312 Regular Session ThB20 Robotics	13:30-15:30 Orchid Junior 4311 Regular Session ThB21 Predictive Control for Nonlinear Systems II
16:00-18:00 Roselle Junior 4711 Regular Session ThC12 Centralized Control	16:00-18:00 Roselle Junior 4613 Regular Session ThC13 Networked Control Systems III	16:00-18:00 Roselle Junior 4612 Regular Session ThC14 Nonlinear System Identification	16:00-18:00 Roselle Junior 4611 Regular Session ThC15 Robust Control II	16:00-18:00 Peony Junior 4512 Regular Session ThC16 Smart Grid I	16:00-18:00 Peony Junior 4511 Regular Session ThC17 Iterative Learning Control II	16:00-18:00 Peony Junior 4412 Regular Session ThC18 Stability of Nonlinear Systems II	16:00-18:00 Peony Junior 4411 Regular Session ThC19 Predictive Control for Linear Systems I	16:00-18:00 Orchid Junior 4312 Regular Session ThC20 Robotics and Nonholonomic Systems	16:00-18:00 Orchid Junior 4311 Regular Session ThC21 Predictive Control for Nonlinear Systems III



Applications

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Click on the day's program and use the arrow keys for easy horizontal and vertical scrolling

CDC 2023 Technical Program Friday December 15, 2023									
Track 12	Track 13	Track 14	Track 15	Track 16	Track 17	Track 18	Track 19	Track 20	Track 21
08:30-09:30 Roselle Simpor Main Ballroom 4601AB-4806 Plenary Session FrP1 Machine Learning: Bane or Boon for Control?									
10:00-12:00 Peony Junior 4412 Regular Session FrA12 Adaptive Control I	10:00-12:00 Roselle Junior 4613 Regular Session FrA13 Networked Control Systems IV	10:00-12:00 Roselle Junior 4612 Regular Session FrA14 Observers for Linear Systems	10:00-12:00 Roselle Junior 4611 Regular Session FrA15 Robust Control III	10:00-12:00 Peony Junior 4512 Regular Session FrA16 Smart Grid II	10:00-12:00 Peony Junior 4511 Regular Session FrA17 Statistical Learning I	10:00-12:00 Peony Junior 4412 Regular Session FrA18 Stability of Nonlinear Systems III	10:00-12:00 Peony Junior 4411 Regular Session FrA19 Predictive Control for Linear Systems II	10:00-12:00 Orchid Junior 4312 Regular Session FrA20 Modelling and Control I	10:00-12:00 Orchid Junior 4311 Regular Session FrA21 Extremum-Seeking Control
13:30-15:30 Peony Junior 4412 Regular Session FrB12 Adaptive Control II	13:30-15:30 Roselle Junior 4613 Regular Session FrB13 Networked Systems	13:30-15:30 Roselle Junior 4612 Regular Session FrB14 Observers for Nonlinear Systems I	13:30-15:30 Roselle Junior 4611 Regular Session FrB15 Resilient Control Systems	13:30-15:30 Peony Junior 4512 Regular Session FrB16 Emerging Control Applications	13:30-15:30 Peony Junior 4511 Regular Session FrB17 Statistical Learning II	13:30-15:30 Peony Junior 4412 Regular Session FrB18 Stability of Nonlinear Systems IV	13:30-15:30 Peony Junior 4411 Regular Session FrB19 Filtering	13:30-15:30 Orchid Junior 4312 Regular Session FrB20 Modelling and Control II	13:30-15:30 Orchid Junior 4311 Regular Session FrB21 Transportation Networks
16:00-18:00 Peony Junior 4412 Regular Session FrC12 Adaptive Control III	16:00-18:00 Roselle Junior 4613 Regular Session FrC13 Sensor Networks	16:00-18:00 Roselle Junior 4612 Regular Session FrC14 Observers for Nonlinear Systems II	16:00-18:00 Roselle Junior 4611 Regular Session FrC15 Uncertain Systems	16:00-18:00 Peony Junior 4512 Regular Session FrC16 Control Applications	16:00-18:00 Peony Junior 4511 Regular Session FrC17 Intelligent Systems	16:00-18:00 Peony Junior 4412 Regular Session FrC18 Lyapunov Methods	16:00-18:00 Peony Junior 4411 Regular Session FrC19 Kalman Filtering	16:00-18:00 Orchid Junior 4312 Invited Session FrC20 Mechatronics	16:00-18:00 Orchid Junior 4311 Regular Session FrC21 Aerospace Systems

IEEE CDC Workshop 2023

[Home](#)[Program](#)[Speakers](#)[Registration](#)

The 62nd IEEE Conference on Decision and Control, December 12, Marina Bay Sands, Singapore

Overview

As a major methodology for handling parametric uncertainties in dynamical systems, adaptive control and estimation approaches have attracted much attention in both academia and industry over the past few decades. The classical adaptive control and estimation impose appropriate structural knowledge on parametric uncertainties and achieves only asymptotic error convergence with weak robustness in the absence of a stringent condition termed persistent excitation, which prevent it from widespread applications in real-world systems. In recent years, some advanced adaptive design concepts have been proposed to overcome the above limitations, where notable ones with great potential in practice include regressor extension, online optimization, and non-Euclidean adaptation. These approaches have resulted in several successful real-world applications, but they are limited to relatively simple systems with low degrees of freedom, and in-depth considerations about application issues are rare.



Workshop Program

9:00am **Opening remarks**

Registration and Coffee & Tea!

9:20am **Adaptive Control in Reaction Systems**

Denis Dochain (Universite Catholique de Louvain)

9:55am **Learning Through Lyapunov-based Deep Neural Networks (Lb-DNN)**

Warren Dixon (University of Florida)

10:30am **Coffee & Tea Break**

10:50am **Box-Draper Recursive Estimation for Multi-Response Dynamical Systems**

Martin Guay (Queen's University)

11:25am **The Extended Kalman Filter Revisited**

Robert Mahony (The Australian National University)

12:00pm **Lunch Break**



Applications

- 1:30pm **The Dynamic Adaptation Gain/Learning Rate: An Efficient Solution for Improving Adaptation/Learning Transients**
Ioan Dore Landau (French National Centre for Scientific Research)
- 2:05pm **Conquering Time-Varying Parameters in Adaptive Control: Parameter Congelation and Active Nodes**
Alessandro Astolfi (Imperial College London)
- 2:40am **Coffee & Tea Break**
- 3:00pm **Composite Learning Robot Control**
Yongping Pan (Sun Yat-sen University)
- 3:35pm **Recent Results on Parameter Estimation-Based Observers**
Bowen Yi (Polytechnique Montreal)



Adaptive Control v.s. Robust Control

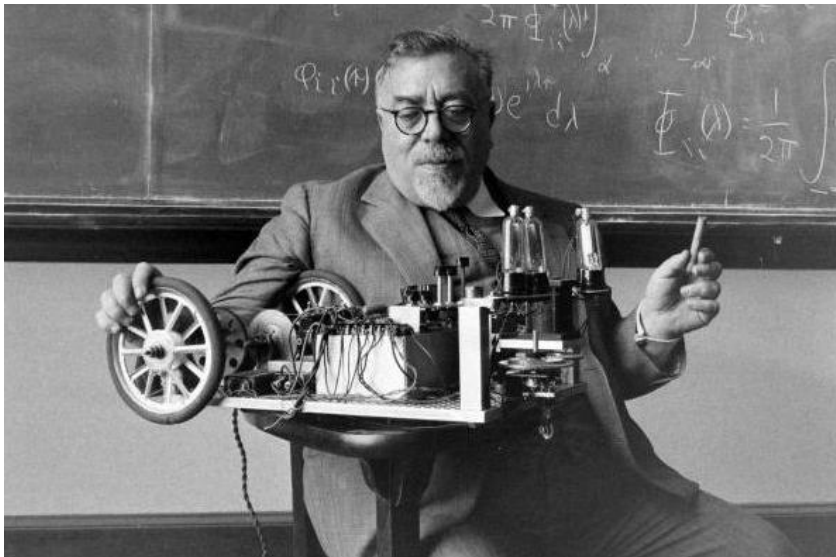
Adaptive Control

- Adaptive control is superior to robust control in dealing with uncertainties in **constant or slow-varying parameters**
- An adaptive controller improves its performance as adaptation goes on
- An adaptive controller requires little or no a priori information about the unknown parameters

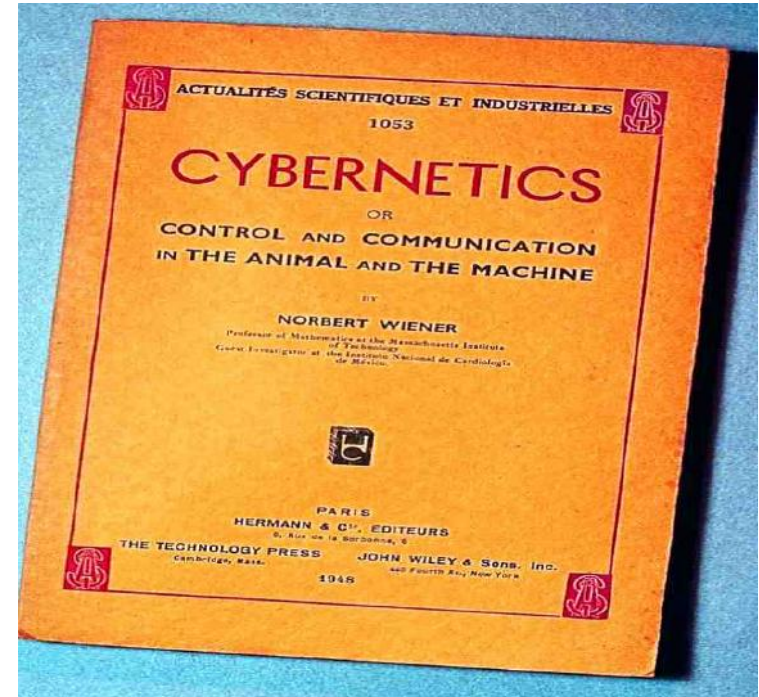
Robust Control

- Robust control has advantages in dealing with **disturbances, quickly varying parameters, and unmodeled dynamics**
- A robust controller attempts to keep consistent performance
- A robust controller requires reasonable a priori estimates of the parameter bounds

Solution: Adaptive **Robust Adaptive Control** line controller



Norbert Wiener
(1894.11.26—1964.3.18)
Ph. D. from Harvard
Professor at MIT



《Cybernetics》, 1948



Historical Development

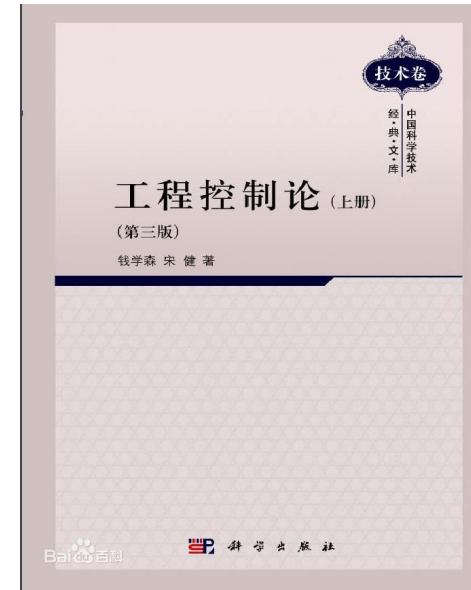


钱学森

(1911.12.11 – 2009.10.31)

Ph. D. from Caltech

Professor at MIT



《工程控制论》, 1953



控制论是关于**工程技术**领域各个系统自动控制和自动调节的理论。**维纳**博士四十年代提出了控制论的基本思想后，不少工程师和数学博士曾寻找通往这座理论顶峰的道路，但均半途而废。工程师偏重于实践，解决具体问题，不善于上升到理论高度，数学家则擅长理论分析，却不善于从一般到个别去解决实际问题。**钱学森**则集中两个优势于一身，高超地将两只轮子装到一辆战车上，碾出了工程控制论研究的一条新途径。



从科学理论的角度来看，20世纪上半叶的三大伟绩是相对论、量子论和控制论，也许可以称它们为三项科学革命，是人类认识客观世界的三大飞越。



Historical Development

- Prior to 1940: Phase plane method introduced by Poincare in mathematica
- Wartime problems: phase plane method and describing function method
- 1950s-1960s: adaptive control-gain scheduling, MIT rule
- Interest diminished due to the crash of the X-15A-3

Crash of NASA X-15A-3

- The crash of the X-15A-3 (November 15, 1967)



On Nov. 15, 1967, on Michael Adams seventh flight, he entered a spin from which he was able to recover but could not bring it out of an inverted dive because of a technical problem with **the adaptive flight control system**.

Crash due to stable, albeit non-robust adaptive controller!

Crash site of the X-15A-3



Adaptive Control and the NASA X-15-3 Flight Revisited



NASA CRYSTAL FLIGHT RESEARCH CENTER (NASA CFCR)

LESSONS LEARNED AND LYAPUNOV-STABILITY-BASED DESIGN

Decades after the first hypersonic vehicles pushed the boundaries of aerospace technology and research, high-performance aircraft continue to be the subject of considerable research interest. A new generation of hypersonic vehicles offers a far more effective way of launching small satellites or other vehicles into low-Earth orbit than expendable rockets. Additionally, these aircraft facilitate quick response and global strike capabilities. High-performance missions involving the X-15 in the 1950s and 1960s and the X-43A in the 2000s pushed the boundaries of aircraft speed and altitude, setting world records. These aircraft also served as platforms for cutting-edge research in propul-

sion, hypersonic stability, and control, as well as supporting technologies that enabled the design and operation of subsequent aircraft and spacecraft. This article examines the role of control in NASA's X-15 program and, in particular, the X-15-3, which used adaptive control. The X-15 aircraft is depicted in Figure 1.

ZACHARY T. DYDEK,
ANURADHA M. ANNASWAMY,
and EUGENE LAVRETSKY

Control of hypersonic vehicles is challenging due to the changes in the aircraft dynamics as the maneuver takes the aircraft over large flight envelopes. Three hypersonic planes, the X-15-1, X-15-2, and X-15-3, were flown as a part of the NASA X-15 program. The X-15-1 and X-15-2 were equipped with a fixed-gain stability augmentation system. In contrast, the X-15-3 was one of the earliest aircraft to feature an adaptive control scheme. The Honeywell MH-96 self-adaptive controller adjusted control

In summary, a sudden change in actuator effectiveness, which could have been caused by the electrical disturbance, causes the dynamics to depart significantly from those represented in the model and therefore in the control design. As a result, the control gain choices, despite the flexibility provided by the adaptive feature, are inadequate, causing the overall control system to be unable to recover from the onset of instability leading up to the crash.

The MH-96 adaptive flight control system is an elegant design that accomplished its goal of enforcing performance across all flight conditions. Furthermore, the MH-96 showed that a satisfactory adaptive control system could be designed without having accurate a priori information about the aircraft aerodynamics, and, consequently, aircraft configuration changes could be easily accounted for [40]. However, the MH-96 lacked an analytically based proof of stability, which was highlighted by the fatal crash in 1967. After four decades, the theoretical ground work for applying adaptive control has now made it possible to design adaptive controllers that offer high performance as well as stability guarantees in the presence of uncertainties.

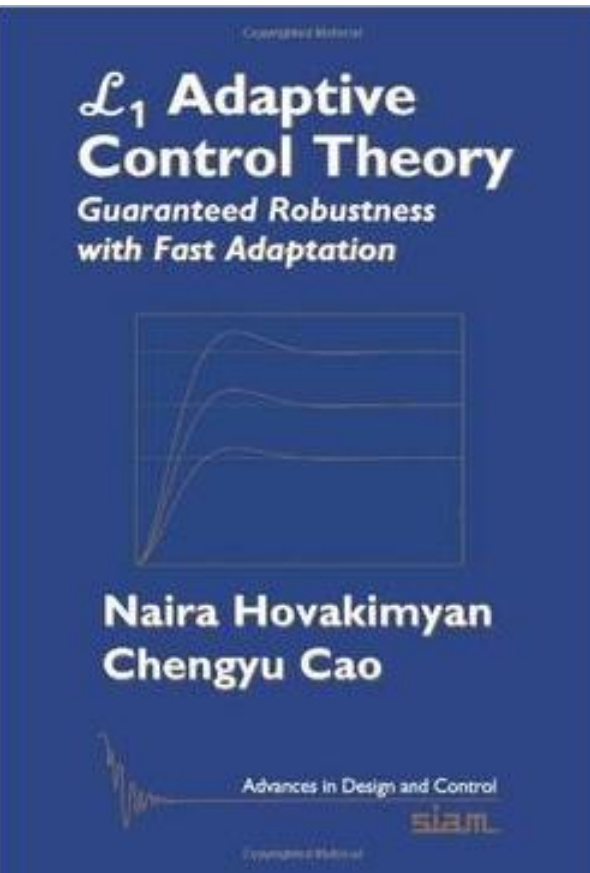
Digital Object Identifier 10.1109/MCS.2010.934292

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➤ Mid 1950s: Autopilots for high-performance aircrafts (solved by gain scheduling)



IEEE TRANSACTIONS ON AUTOMATIC CONTROL, VOL. 59, NO. 11, NOVEMBER 2014

3075

L_1 -Adaptive Control: Stability, Robustness, and Interpretations

P. A. Ioannou, A. M. Annaswamy, K. S. Narendra, S. Jafari, L. Rudd, R. Ortega, and J. Boskovic

Abstract—An adaptive control design approach that involves the insertion of a strictly proper stable filter at the input of standard Model Reference Adaptive Control (MRAC) schemes has been proposed in the recent years. This approach was given the name L_1 -Adaptive Control (L_1 -AC) due to the L_1 bounds obtained for various signals. As part of the approach it is recommended to use very high adaptive gains for fast and robust adaptation. The purpose of this note is to analyze whether L_1 -AC provides any improvements to existing MRAC schemes by focusing on a simple plant whose states are available for measurement presented in [1]. Our analysis shows that the insertion of the proposed filter deteriorates the performance and robust stability margin bounds compared to standard MRAC, i.e., when the filter is removed. The use of high adaptive gains recommended in the L_1 -AC approach may cause two major problems. First, it makes the differential equation of the adaptive law very stiff leading to possible numerical instabilities. Second, it makes the adaptive scheme less robust with respect to unmodeled dynamics.

Index Terms— Model reference adaptive control (MRAC).

The purpose of this note is to discuss the possible benefits that the L_1 -AC approach was intended to achieve. We focus on a simple plant whose states are available for measurement presented in [1]. The conclusions derived are applicable to all other results on L_1 -AC that involve insertion of strictly proper stable filters at the input of MRAC schemes for different plant models and plant assumptions. The reason is that the insertion of any strictly proper stable filter at the input of MRAC changes the relative degree of the plant and violates the necessary condition for tracking the output of the reference model at all frequencies [8]–[19]. The methodology used in this note in evaluating the L_1 -AC approach involves the following steps: Identify the control objective; examine whether it can be met and how closely; and evaluate whether there are any improvements, i.e., what is gained by the filter so that when it is removed we can see a deterioration.

It is evident in [1] and other L_1 -AC publications [2]–[5] that the control objective is not clear and that there is no adequate discussion as to how it is achieved despite the numerous bounds. For example, we clarify in this note that in [1] and other publications [2]–[5], there are two control objectives; one that aims to track the reference input of the reference model and one that aims to track the output of the reference model. In addition to the above observation, the note derives

➤ 2020s: Meta Adaptive Nonlinear Control



Thank You !